

Irish Math Olympiad 1992 — P6/10

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Let $n > 2$ be an integer and let $m = \sum k^3$, where the sum is taken over all integers k with $1 \leq k < n$ that are relatively prime to n . Prove that n divides m . (Note that two integers are *relatively prime* if, and only if, their greatest common divisor equals 1.)

Solution

Let q be a positive integer that is coprime with n . Due to Euclid's algorithm $n - q$ is also coprime with n . Thus if we consider the sum modulo n , when $k = q$ we know that q^3 is contributed to the sum however since $k = n - q$ will also appear we know that $-q^3$ will also be contributed to the sum. In total these values of k have contributed 0 to the sum mod n .

Since we can pair up all of the integers k that are coprime with n like so $(q, n - q)$ and we know that these values of k will contribute 0 to the sum mod n , we conclude that $m \equiv 0 \pmod{n}$, that is $n \mid m$. ■